Satisfactory. Part (a)(i) was well answered. In (a)(ii)(iii), some candidates failed to sketch the variation of the speed of Train A correctly and therefore they were unable to determine the two trains' separation using the area under the graph. Candidates' performance in (b) was fair. In (b)(i), many candidates did not apply the principle of conservation of momentum correctly in the situation concerned. In (b)(ii), only the more able ones identified the correct velocities to be used in finding the rate of change of momentum of Train A.

This question required candidates to describe how to measure the speed of a bullet using the apparatus provided. The general performance was poor. Many candidates failed to mention that the speed of the trolley immediately after the collision should have been taken. A small number of candidates did not know that the motion sensor registered the trolley's speed instead of its distance travelled. Not many were able to state the precautions for getting a more accurate result such as the bullet should be fired along the trolley's direction of travel.

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Part (a)(i) was well answered. In (a)(ii), only the more able candidates understood that as the spring gun was fixed, external force(s) acted on the system and thus momentum was not conserved. Moreover, some candidates only mentioned in their answers that momentum was not the same before and after the process, without giving any explanation. Part (b) was well answered. However, a few candidates had difficulties in resolving the initial velocity of the cannon ball correctly into its vertical and horizontal components. In (c), only the more able ones reasoned without any numerical calculation that t_f would increase as the initial vertical velocity v sin · increased.

Candidates did well in (a) which asked for how to demonstrate the conservation of energy. A few of them forgot to point out the measurements to be made. Part (b) examined candidates' knowledge and understanding of momentum in the context of Newton's cradle. The overall performance was fair. In (b)(i)(I), some candidates mixed up the conservation of linear momentum and the conservation of mechanical energy and stated 'no work done by the tensions' as the explanation. In explaining why the case was impossible in (b)(i)(II), a few candidates did not calculate the initial and final kinetic energy of the system or failed to find the correct values. Quite a number of the candidates failed to account for the loss in kinetic energy in the repeated collisions in (b)(ii).

This question assessed candidates' understanding of experimental design and force analysis during a collision. The performance was poor. While most recognised the action-reaction pair between *A* and *B*, many failed to draw the force-time graphs correctly.

(No Section B candidate-performance note.)